Lesson 9.2 Guided Notes

Session 1 · From Table to Graph

I can turn a table of values into points and plot them on a graph. · 6.EE.C.9

Word / Spanish	Meaning — complete the blank	Example
ordered pair / par ordenado	Two numbers (x, y) that tell where a _____ is.	(3, 12)
coordinates / coordenadas	The _____ numbers in an ordered pair.	x-coordinate 3, y-coordinate 12
axis / eje	One of the two _____ on a graph.	x-axis across, y-axis up
plot / marcar	To _____ a point on a graph.	right 3, up 12

Key idea

Each row of a table becomes an ordered pair: x _____, then y.

To plot (x, y), start at (0, 0). Move _____ x, then _____ y.

Worked example

Plot the table for y = 4x with x = 1, 2, 3.

The rows become (_____, 4), (2, _____) and (3, 12).

Plot (1, 4): right 1, then up _____.

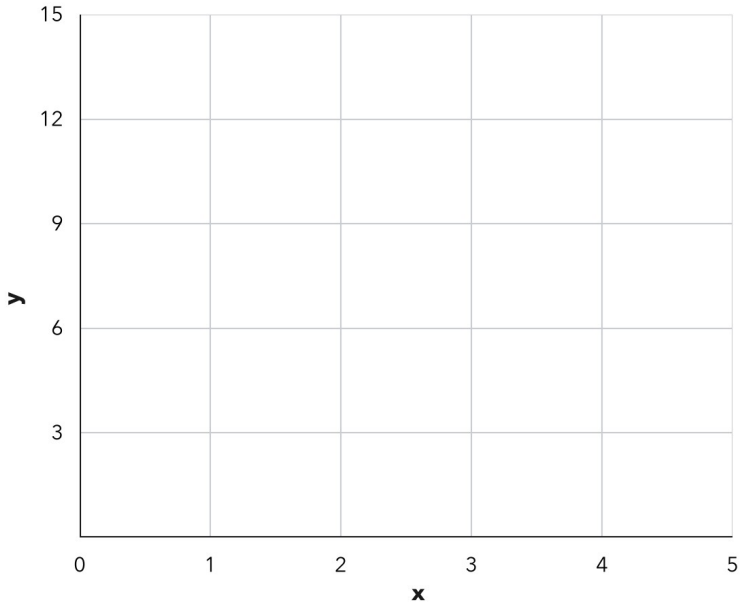
Answer: The points line up in a _____ line.

You try

1. Write the ordered pairs for y = 6x.
a. x = 1 _____
b. x = 2 _____
c. x = 4 _____

_____

2. Plot the points (1, 3), (2, 6) and (4, 12). What shape do they make?



_____

Sum it up: To turn a table into a graph, I ___, and I know the order matters because ___.

_____

Session 2 · Analyzing a Graph

I can read a relationship off a graph and describe the rate it shows. · 6.EE.C.9

Word / Spanish	Meaning — complete the blank	Example
rate of change / tasa de cambio	How much y changes when x goes up by _____.	up $4 per hour
unit rate / tasa unitaria	The amount of y for _____ unit of x.	60 miles per hour
origin / origen	The point _____, where the two axes cross.	(0, 0)
steeper / más empinada	A line that rises _____ as you move right.	up 9 per hour beats up 2

Key idea

Rate of change = change in y _____ change in x.

For a straight-line proportional relationship, use a point with $x \neq 0$ and divide _____ by x for any point on it.

Worked example

A graph of hours (x) and dollars (y) passes through (2, 6) and (5, 15). Find the rate.

Change in y: 15 − 6 = _____ dollars.

Change in x: 5 − 2 = _____ hours.

Divide: 9 ÷ 3 = _____.

Answer: The rate is $_____ per hour.

You try

1. A line passes through (0, 0) and the point. Find the unit rate.
a. (4, 20) _____
b. (6, 42) _____

_____

2. A graph of hours (x) and miles (y) passes through (2, 8) and (5, 20). Find the rate. Then find the miles at 9 hours.

_____

Sum it up: A steeper line means ___ because ___.

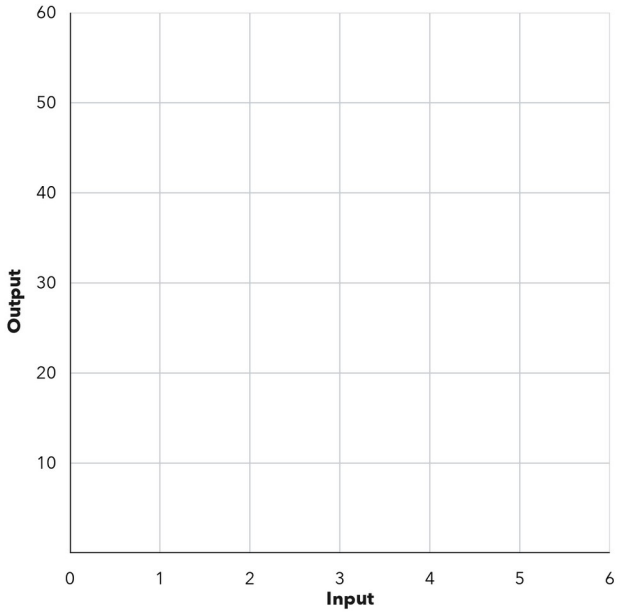
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Session 1 · Model, reason, and explain

From Table to Graph · 6.EE.C.9

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Notebooks cost $4 each, with no fee. Complete a table for x=0,1,3,5, graph the points, and interpret one. Is (2.5,10) a feasible whole-notebook purchase?



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Extend your reasoning

Create a contrasting example that exposes the misconception: connects every intermediate point without considering the context.. Explain what changes and what stays true.

_____

_____

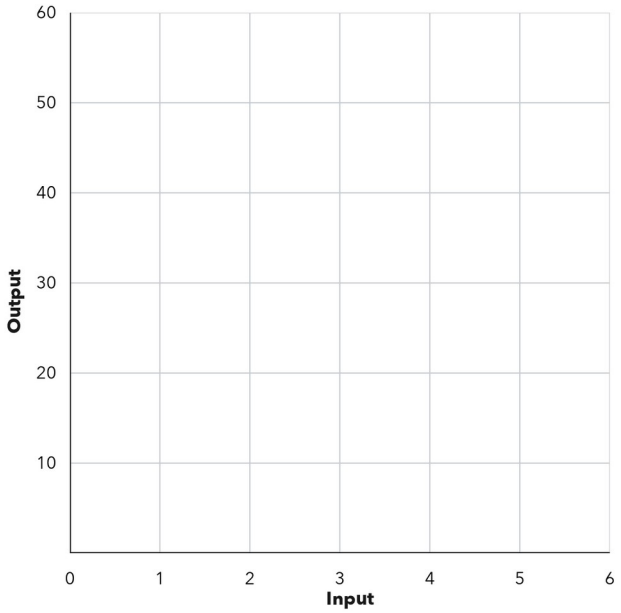
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Analyzing a Graph · 6.EE.C.9

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A rental costs y=3x+5 dollars for x hours. Complete x=0,2,4 and graph. Compare y/x at 2 and 4 with the actual per-hour charge.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: uses y/x for a relationship with a nonzero starting amount.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?